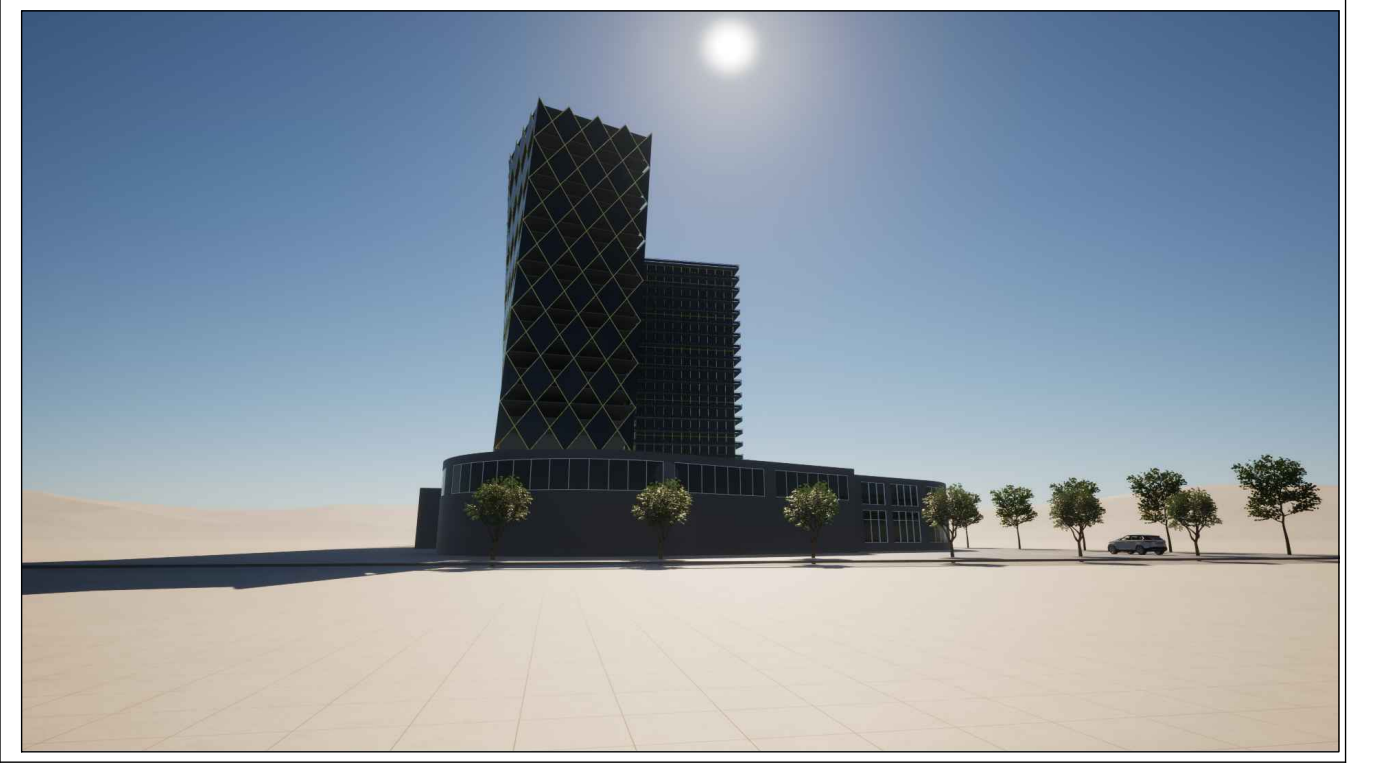
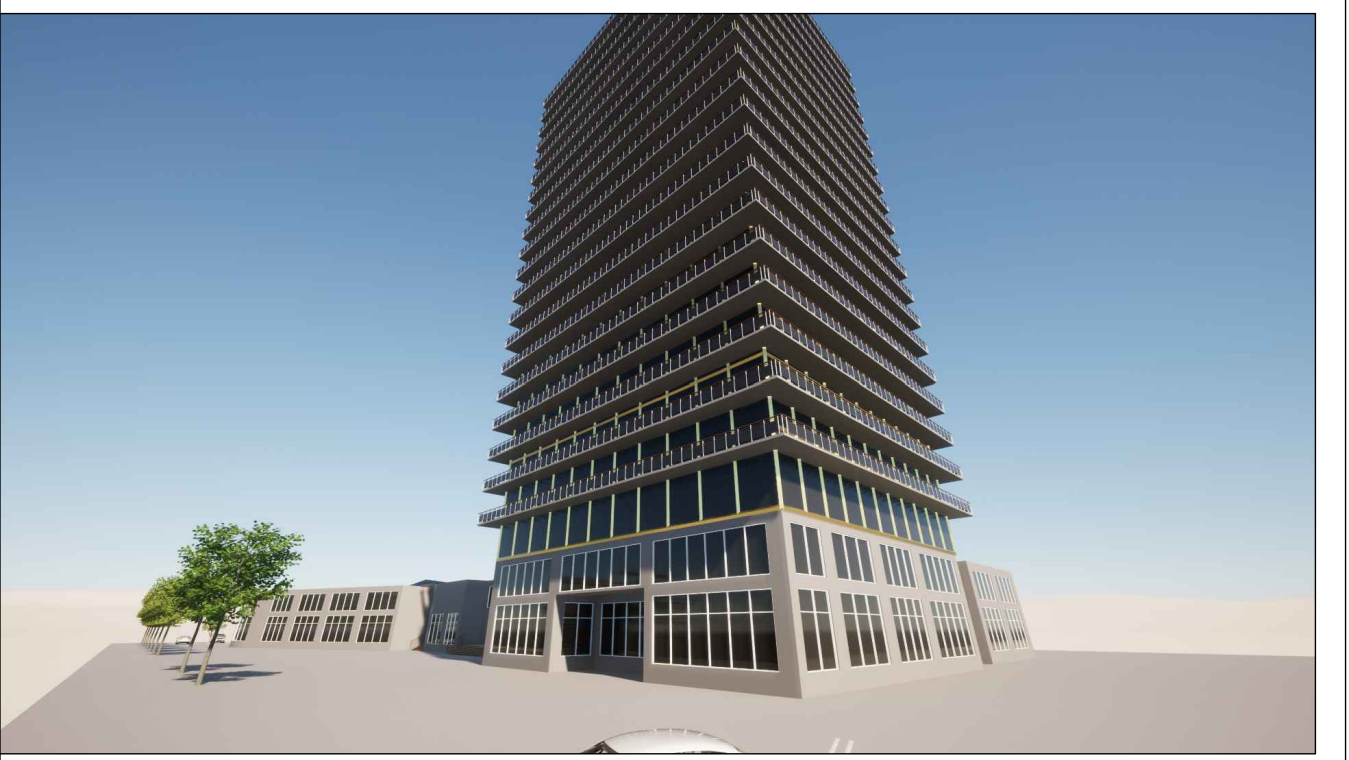
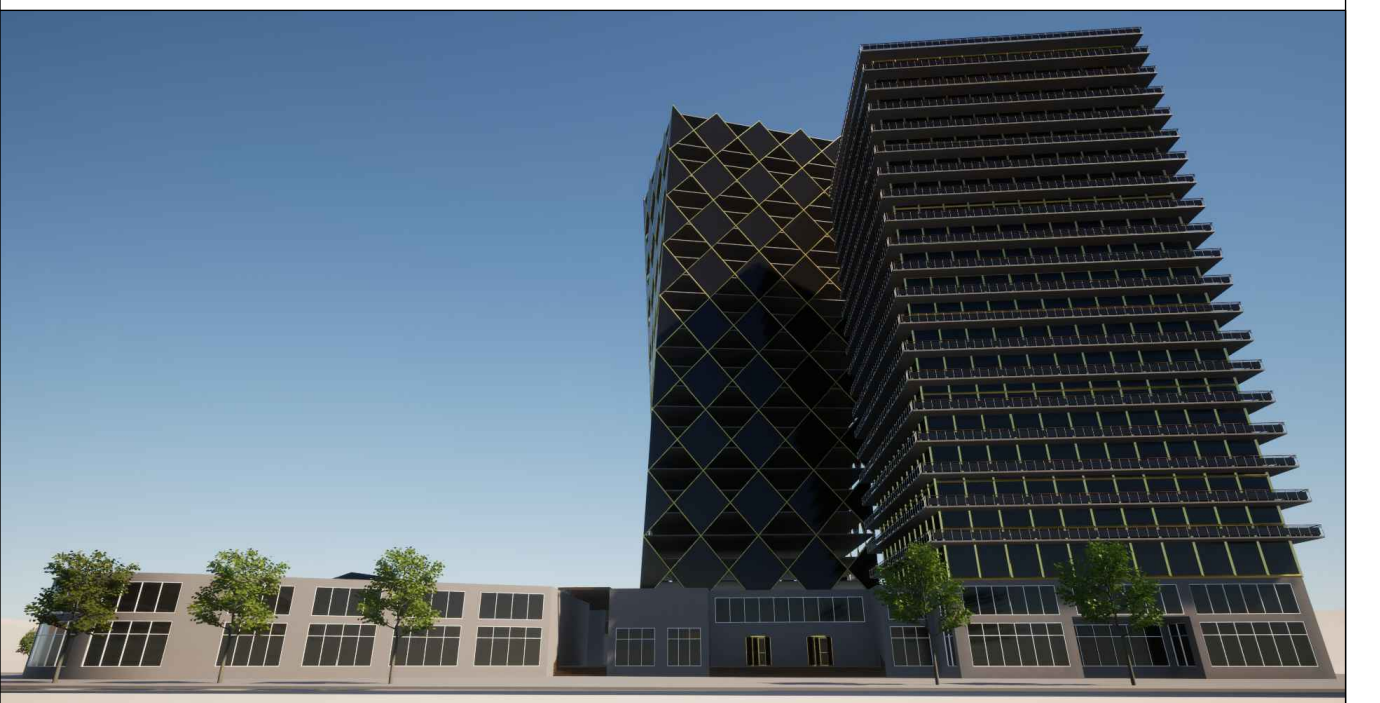
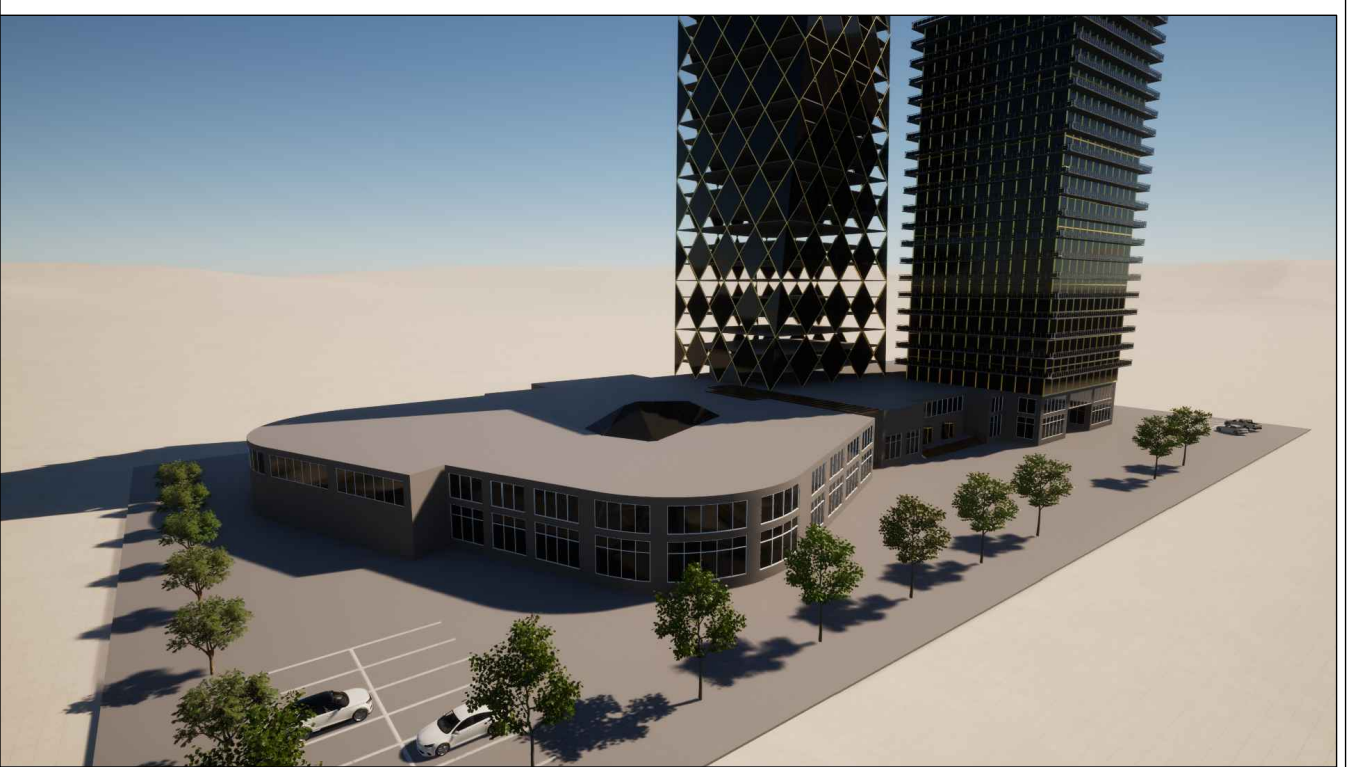
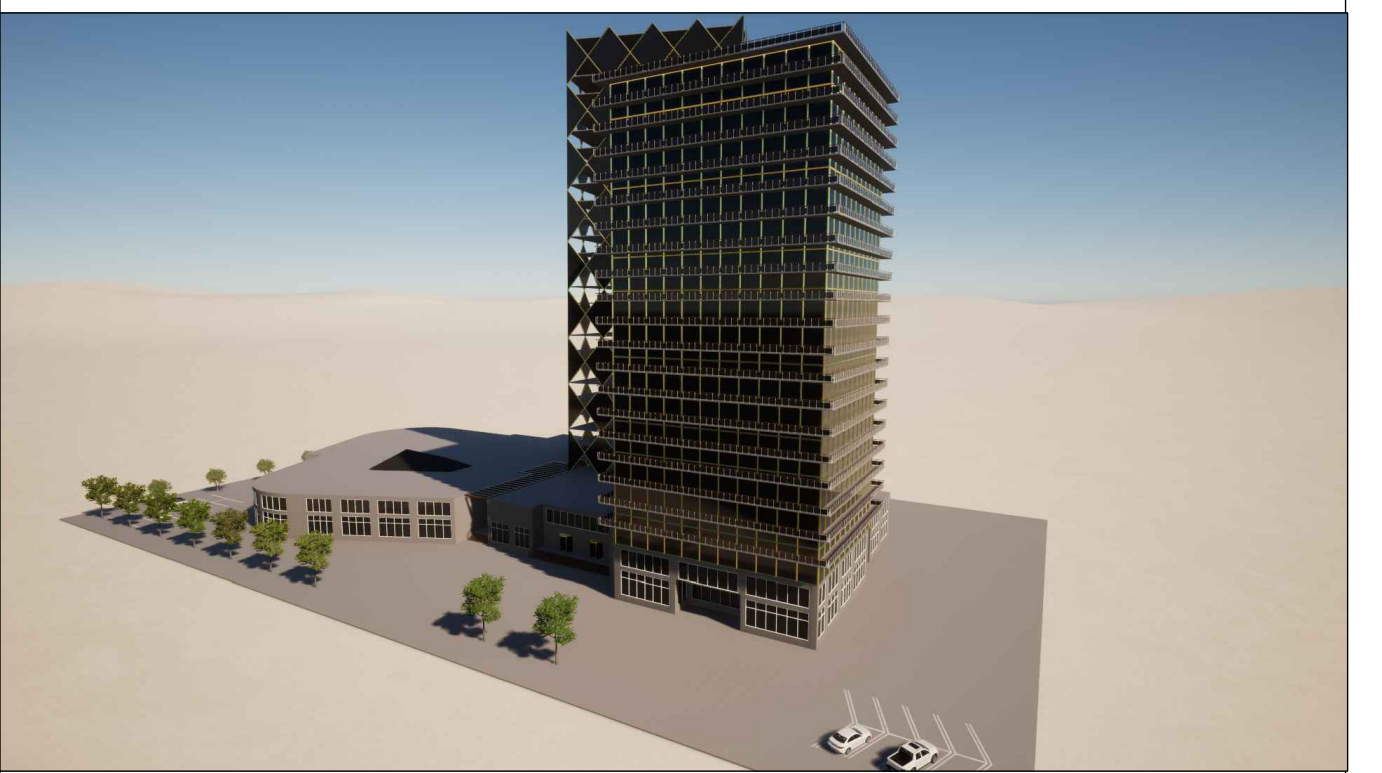
3D SHOTS

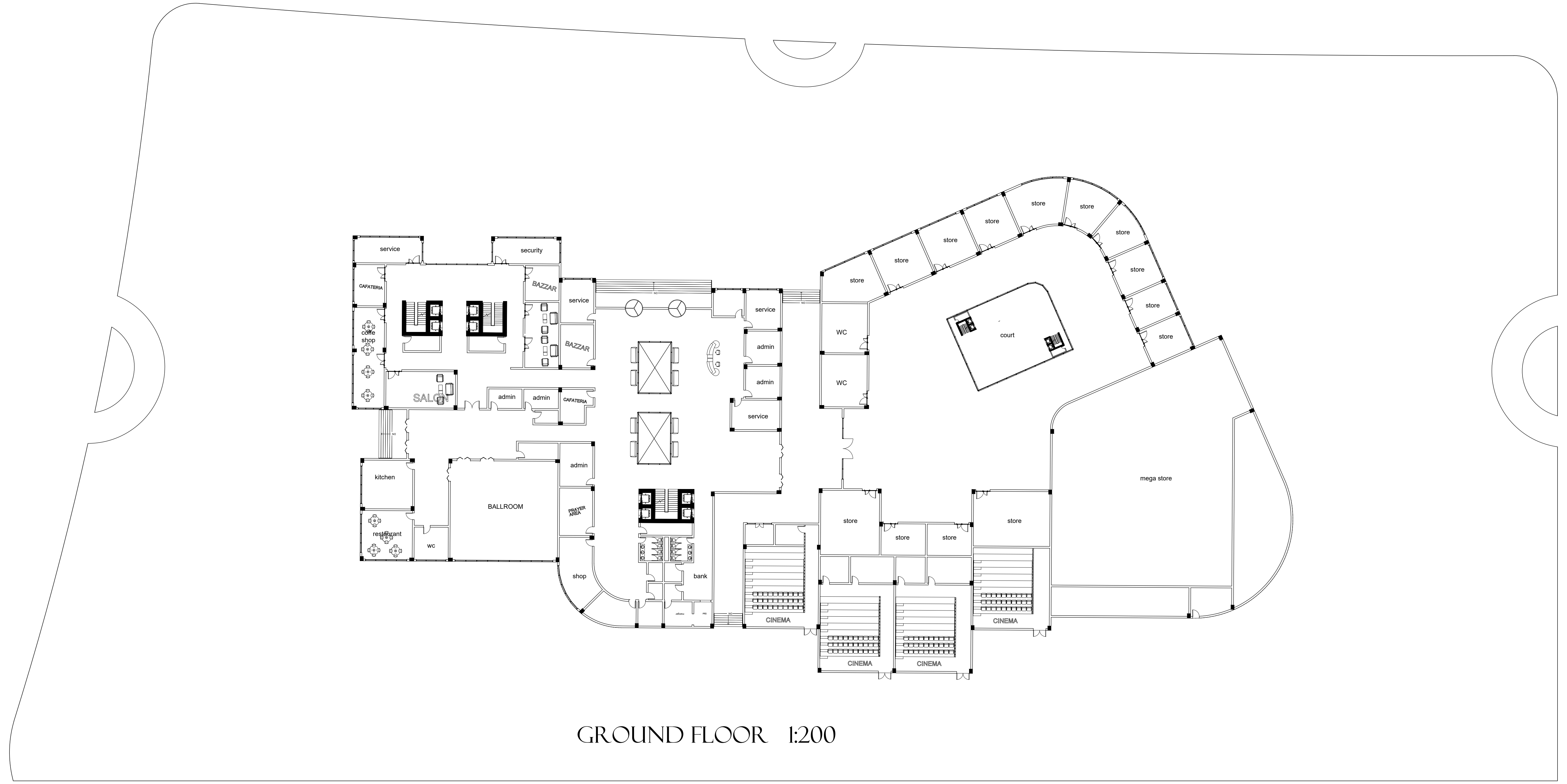








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BAZZAR
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coffe shop
BAZZAR
SALON
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store
store
mega store
kitchen
BALLROOM
admin
PRAYER AREA
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bank
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CINEMA
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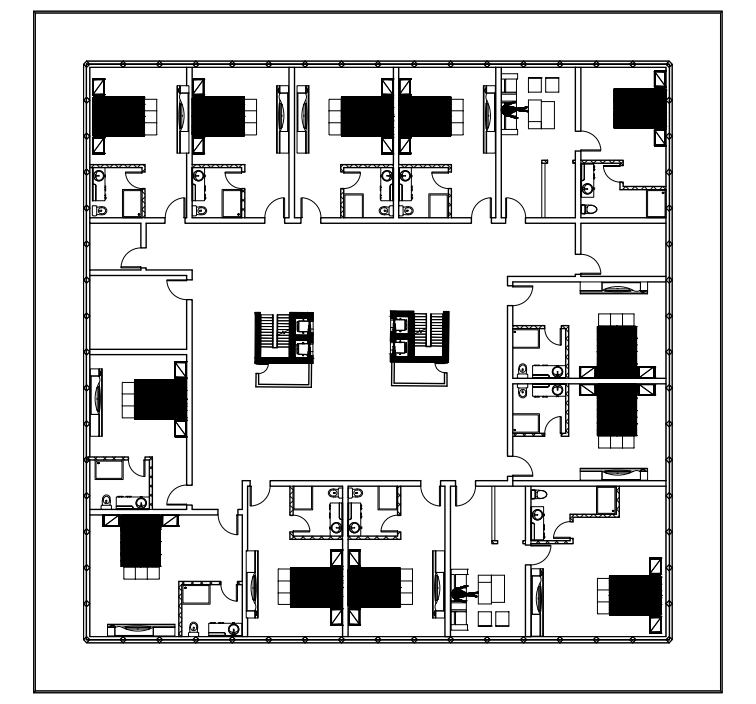
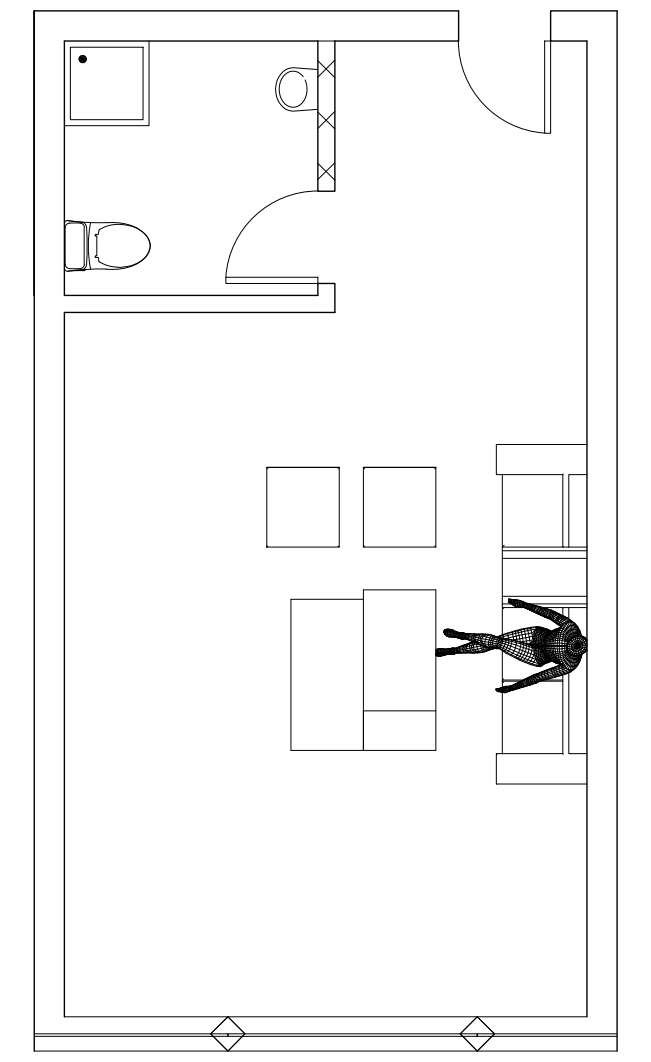
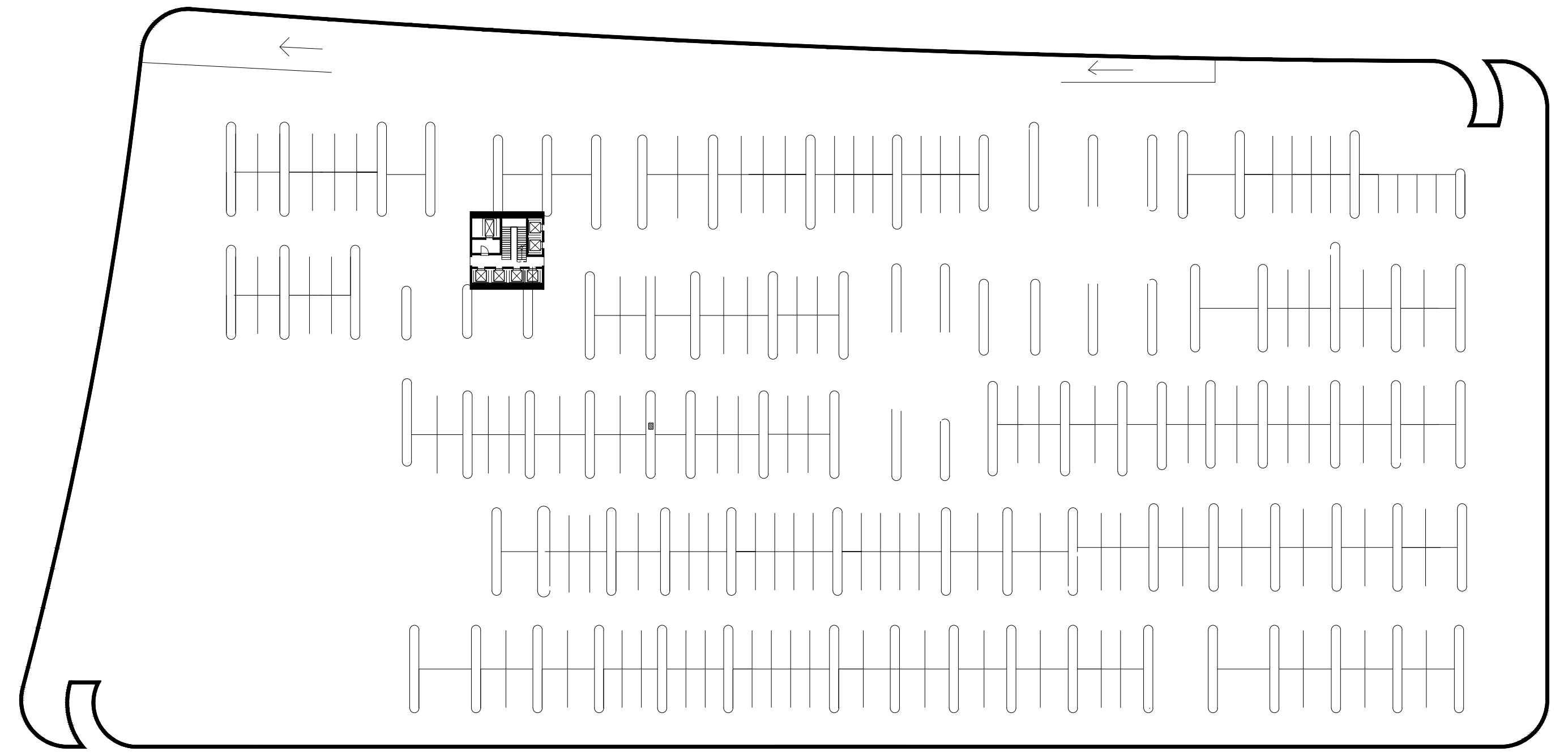
CINEMA

GROUND FLOOR 1:200

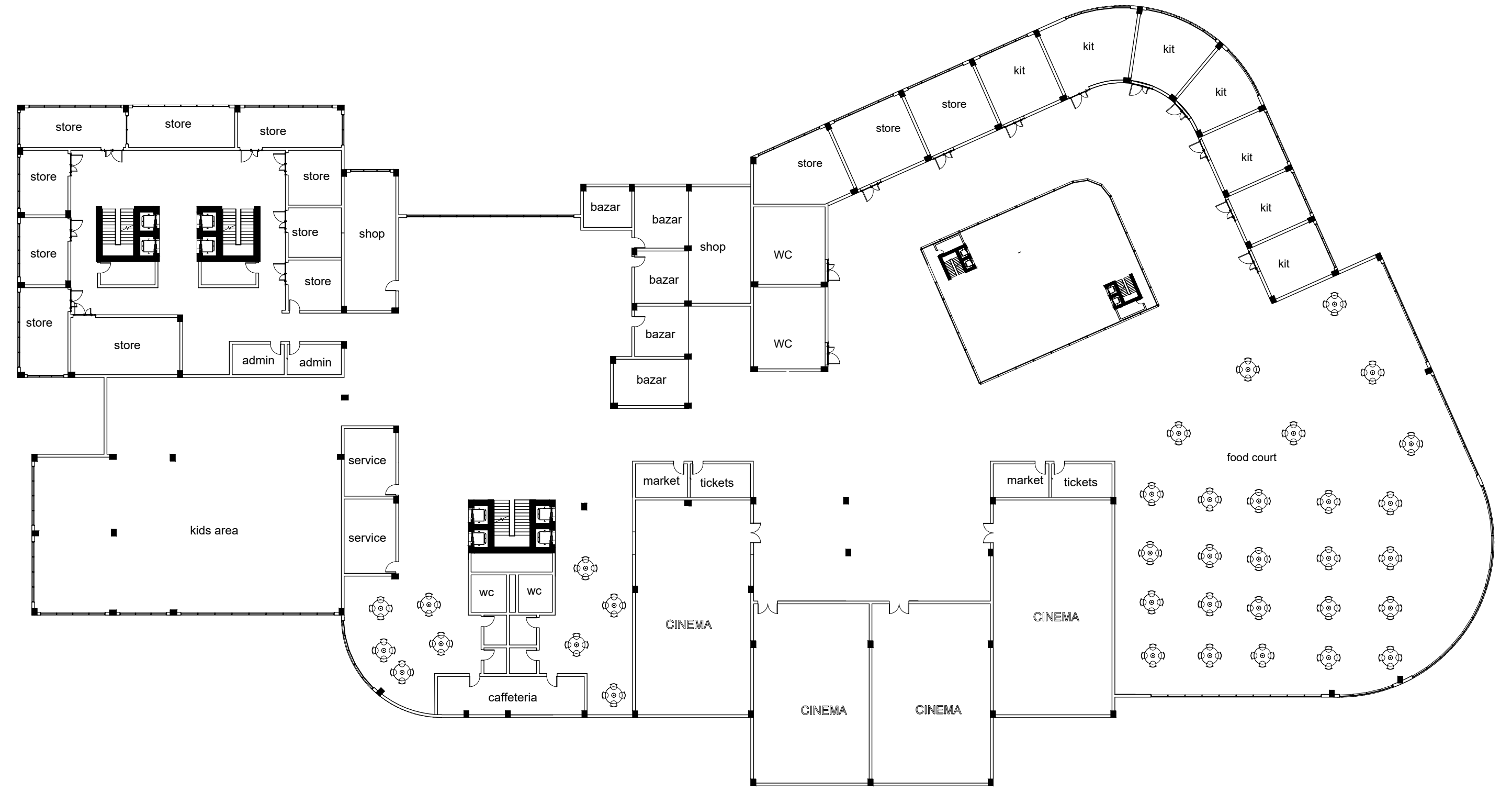


TYPICAL FLOORS
1:400





BASSMENT FLOOR
1:400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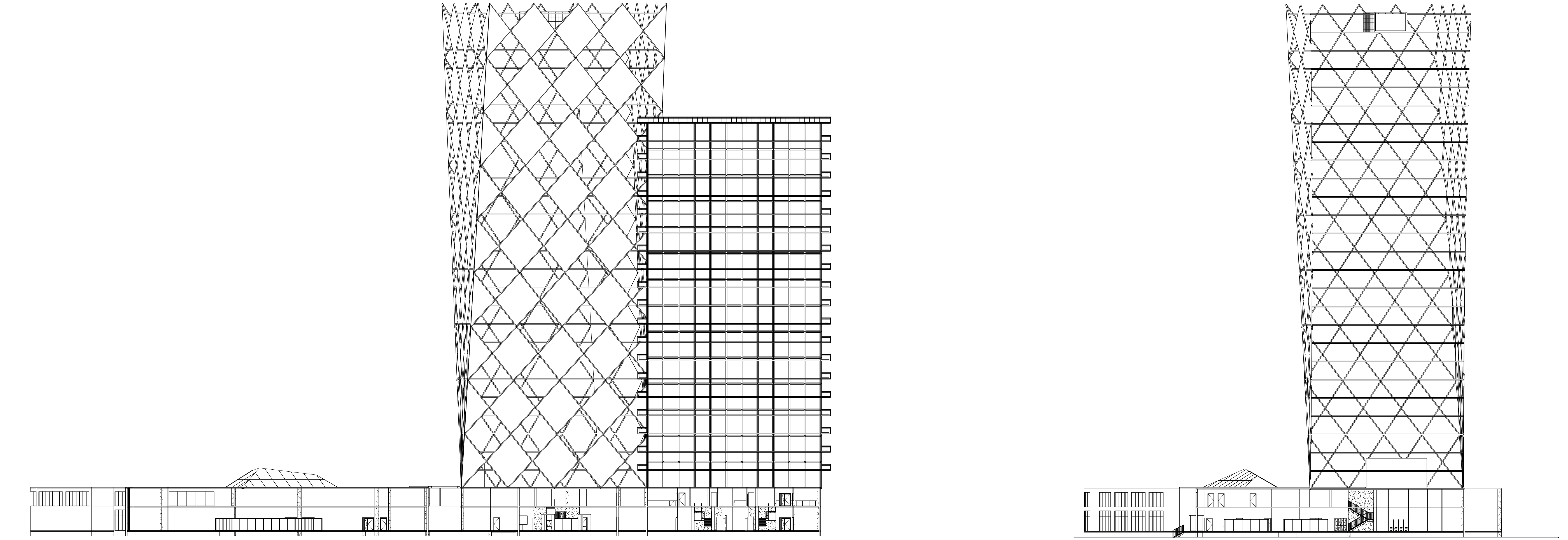
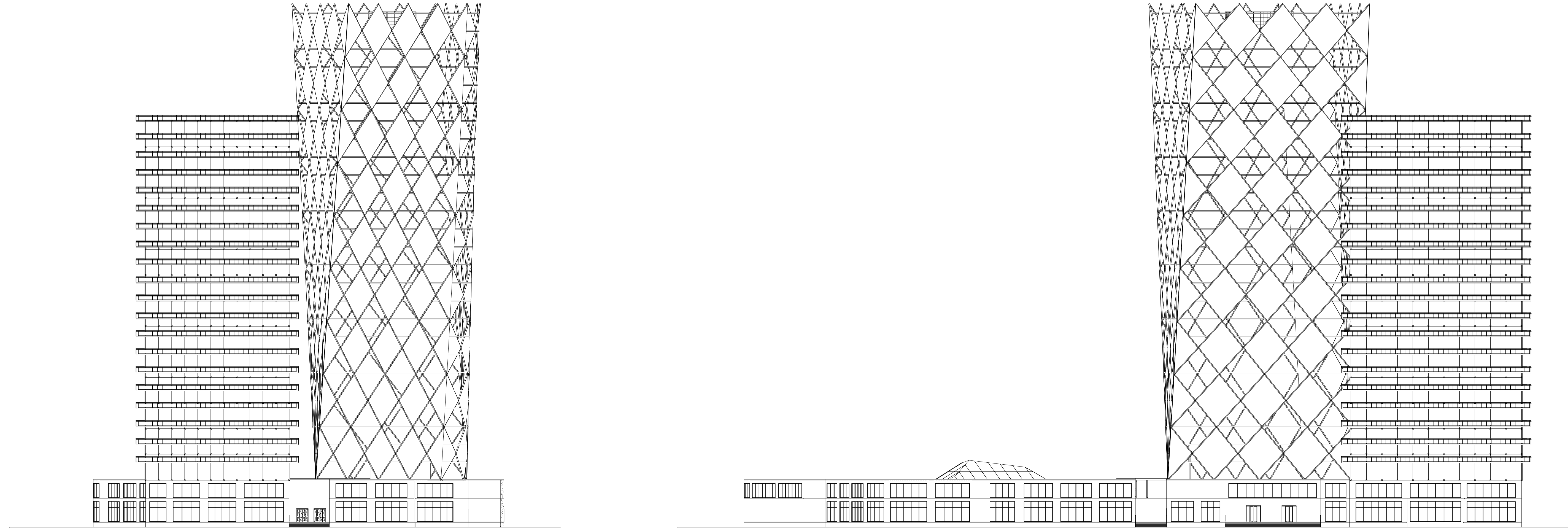
kit
WC
WC
kids area
service
service
wc
wc
caffeteria
market
tickets
market
tickets
food court
CINEMA
CINEMA
CINEMA
CINEMA

FIRST FLOOR 1:400

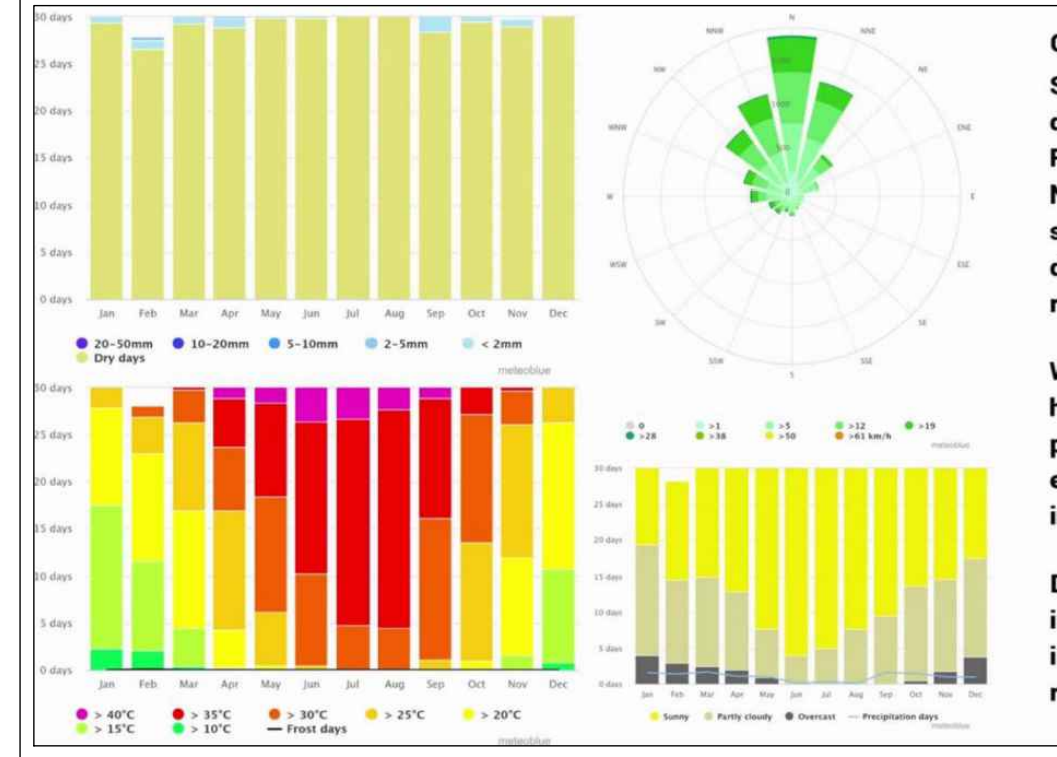


SECTIONS 1:400



ELEVATIONS 1:400

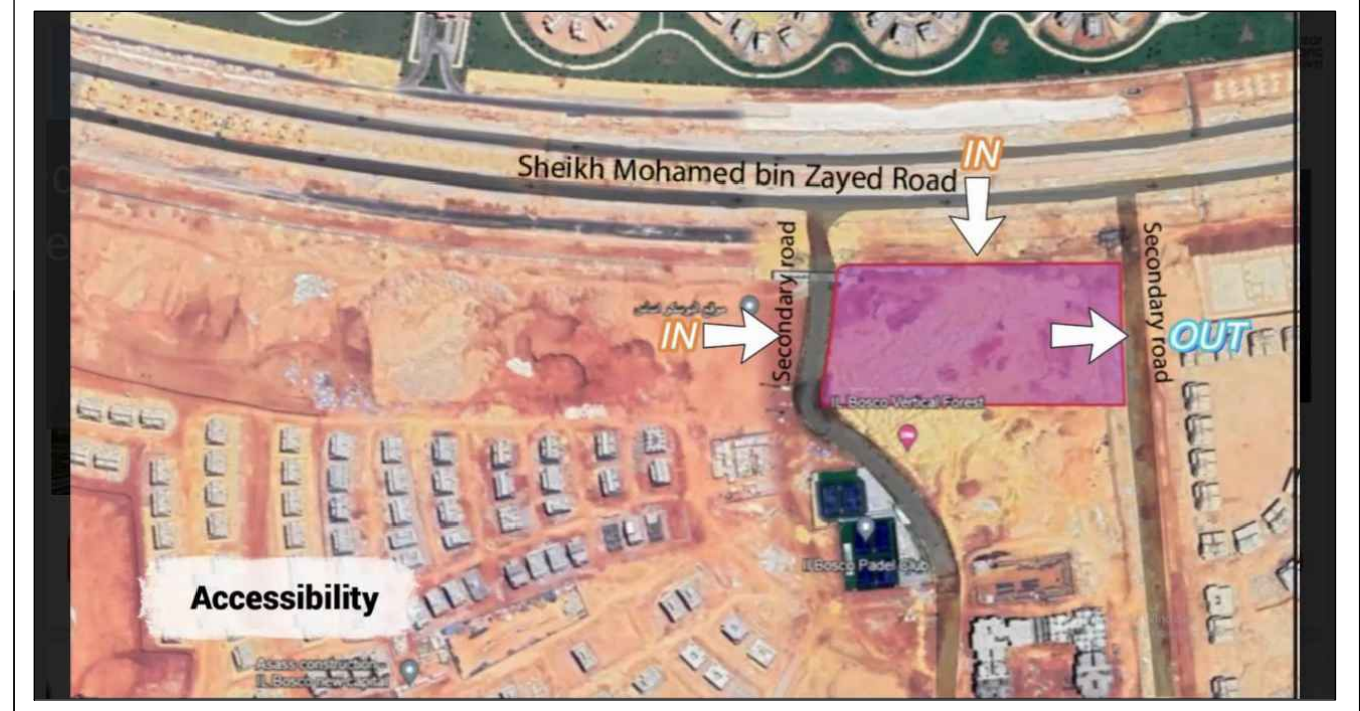
SITE ANALYSIS



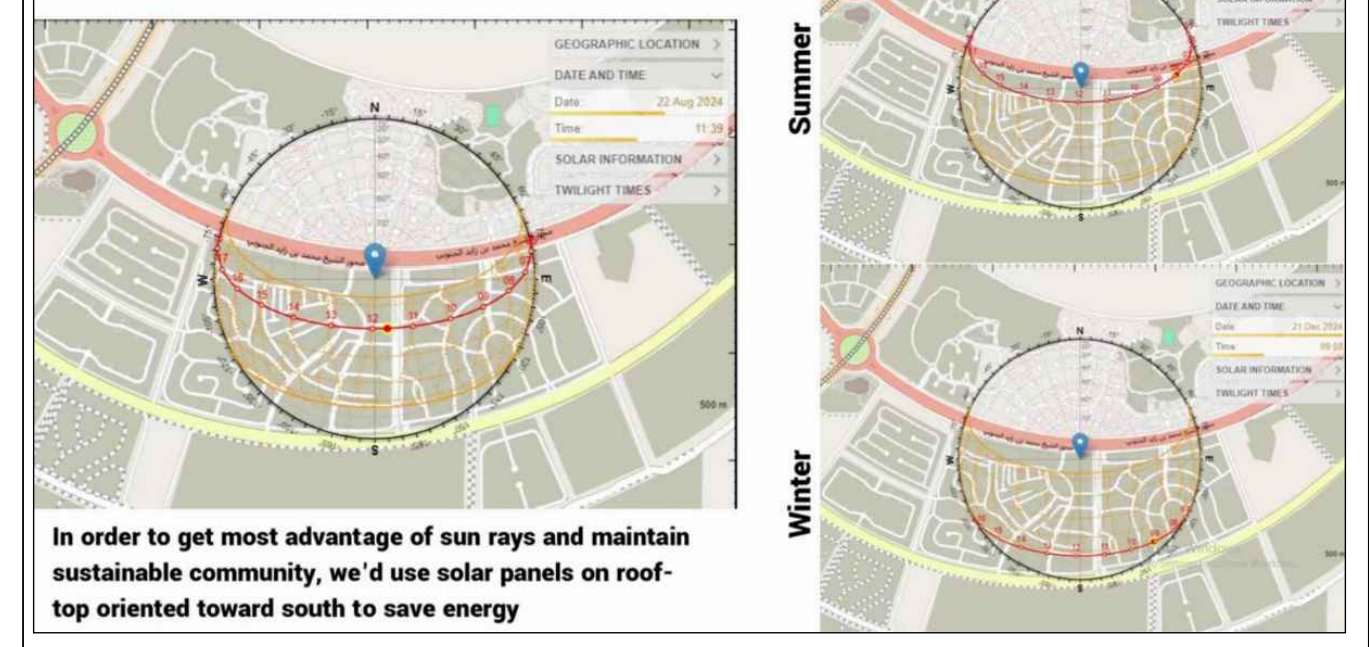
Conclusion:
Since the good wind direction and the green River are located in North direction, it'd be smart architectural descision to orient hotel rooms toward North.

We can benefit from high wind speed and produce renewable and effcient energy by installing wind turbins

Due to high temprature in summer using glass in facade is not recommended.

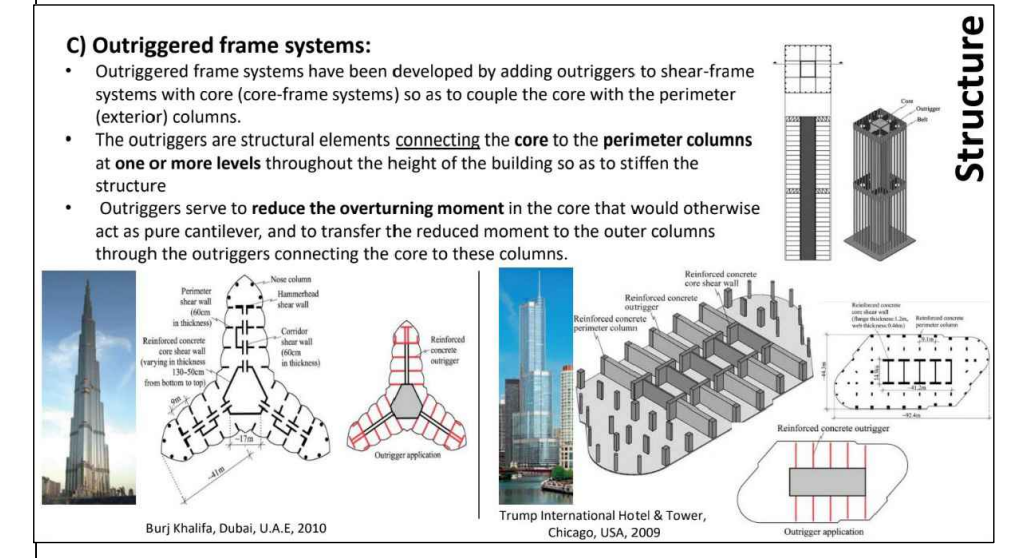

Sheikh Mohamed bin Zayed Road
IN
IN
OUT
Secondary road
Secondary road
Accessibility

Sun Path Diagram


Summer
Winter

In order to get most advantage of sun rays and maintain sustainable community, we'd use solar panels on roof-top oriented toward south to save energy

STRUCTURE SYSTEM

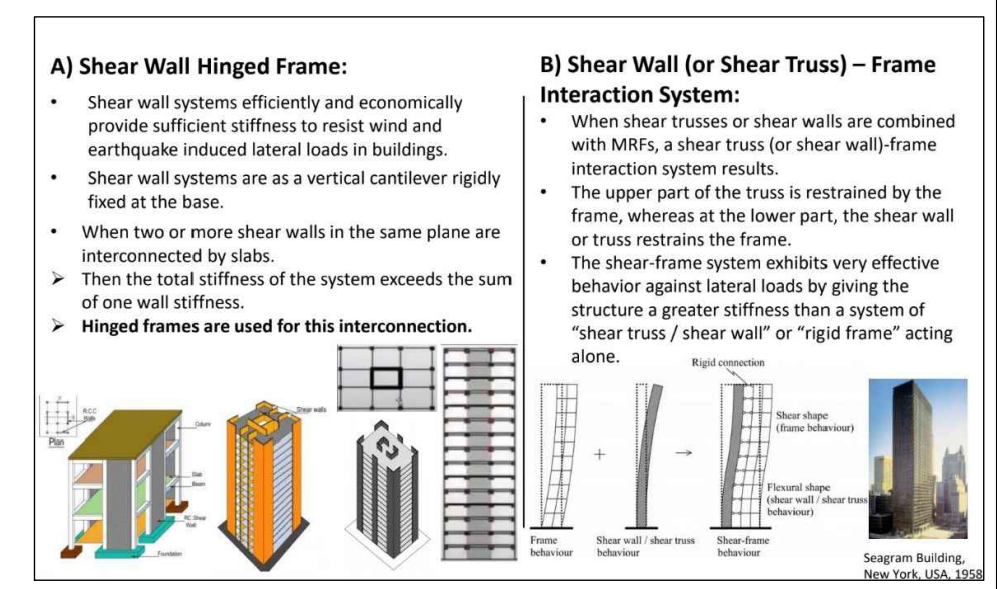


C) Outriggered frame systems:
- Outriggered frame systems have been developed by adding outriggers to shear-frame systems with core (core-frame systems) so as to couple the core with the perimeter (exterior) columns.
- The outriggers are structural elements connecting the **core** to the **perimeter columns** at **one or more levels** throughout the height of the building so as to stiffen the structure
- Outriggers serve to **reduce the overturning moment** in the core that would otherwise act as pure cantilever, and to transfer the reduced moment to the outer columns through the outriggers connecting the core to these columns.

Burj Khalifa, Dubai, U.A.E, 2010

Trump International Hotel & Tower, Chicago, USA, 2009

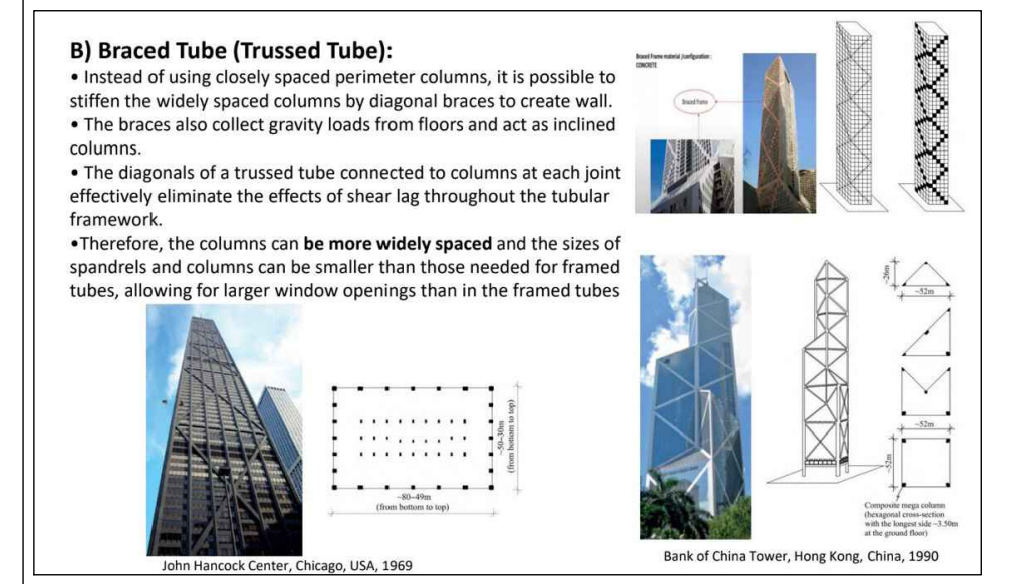
Structure



A) Shear Wall Hinged Frame:
- Shear wall systems efficiently and economically provide sufficient stiffness to resist wind and earthquake induced lateral loads in buildings.
- Shear wall systems are as a vertical cantilever rigidly fixed at the base.
- When two or more shear walls in the same plane are interconnected by slabs.
- Then the total stiffness of the system exceeds the sum of one wall stiffness.
- **Hinged frames are used for this interconnection.**

B) Shear Wall (or Shear Truss) – Frame Interaction System:
- When shear trusses or shear walls are combined with MRFs, a shear truss (or shear wall)-frame interaction system results.
- The upper part of the truss is restrained by the frame, whereas at the lower part, the shear wall or truss restrains the frame.
- The shear-frame system exhibits very effective behavior against lateral loads by giving the structure a greater stiffness than a system of "shear truss / shear wall" or "rigid frame" acting alone.

Seagram Building, New York, USA, 1958



B) Braced Tube (Trussed Tube):
- Instead of using closely spaced perimeter columns, it is possible to stiffen the widely spaced columns by diagonal braces to create wall.
- The braces also collect gravity loads from floors and act as inclined columns.
- The diagonals of a trussed tube connected to columns at each joint effectively eliminate the effects of shear lag throughout the tubular framework.
- Therefore, the columns can **be more widely spaced** and the sizes of spandrels and columns can be smaller than those needed for framed tubes, allowing for larger window openings than in the framed tubes

John Hancock Center, Chicago, USA, 1969

Bank of China Tower, Hong Kong, China, 1990

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